INCH-POUND

MIL-DTL-85891B
11 MAR 2019

SUPERSEDING
MIL-P-85891A
1 APRIL 1992

DETAIL SPECIFICATION

ENGINEERED MEDIA FOR REMOVAL OF ORGANIC COATINGS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

Comments, suggestions, or questions on this document should be addressed to AFLCMC/EZSS, 2145 Monahan Way, Bldg. 28, Wright-Patterson AFB OH, 45433-7017 or e-mailed to ENGINEERING.STANDARDS@US.AF.MIL. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

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1. SCOPE

1.1 Scope.

This specification covers five types of engineered media used for removal of organic coatings in an abrasive blast operation.

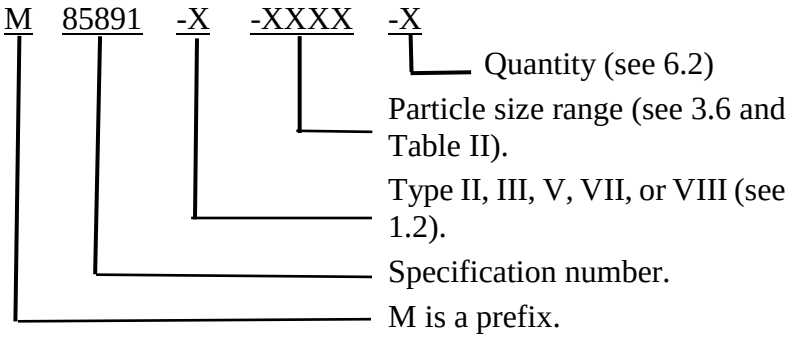
1.2 Classification.

The engineered media are of the following types, as specified (see 6.2):

Type I	DELETED
Type II	Urea Formaldehyde (Thermoset)
Type III	Melamine Formaldehyde (Thermoset)
Type IV	DELETED
Type V	Acrylic (Thermoplastic)
Type VI	DELETED
Type VII	Starch Polymer
Type VIII	Urea/Melamine Multi-Plastic Blends

1.3 Part or identifying number (PIN).

PINs to be used for the plastic media acquired to this specification are created as follows:



2. APPLICABLE DOCUMENTS

2.1 General.

The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks.

The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

CCC-C-440	Cloth, Cheesecloth, Cotton, Bleached and Unbleached

FEDERAL STANDARDS

FED-STD-313	Material Safety Data, Transportation Data, and Disposal Data for Hazardous Materials Furnished to Government Activities

COMMERCIAL ITEM DESCRIPTION

A-A-58054	Abrasive Mats, Non-Woven, Non-Metallic

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-81706	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-PRF-85570	Cleaning Compounds, Aircraft, Exterior

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government documents, drawings, and publications.

The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AIR FORCE TECHNICAL ORDER (TO)

TO 1-1-8	Application and Removal of Organic Coatings, Aerospace and Non-Aerospace Equipment

(Copies of this document are available online at https://www.e-publishing.af.mil.)

NAVAL CIVIL ENGINEERING LABORATORY (NCEL)

NCEL Contract Report (CR) 87.011	Explosibility and Ignitability of Plastic Abrasive Media

(Copies of this document are available online at https://apps.dtic.mil/.)

2.3 Non-Government publications.

The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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ASTM INTERNATIONAL

ASTM D792	Standard Test Methods for Density and Specific Gravity (Relative Density) of Plastics by Displacement
ASTM D1921	Standard Test Methods for Particle Size (Sieve Analysis) of Plastic Materials
ASTM D2240	Standard Test Method for Rubber Property - Durometer Hardness
ASTM D2583	Standard Test Method for Indentation Hardness of Rigid Plastics by Means of a Barcol Impressor
ASTM D3682	Standard Test Method for Major and Minor Elements in Combustion Residues from Coal Utilization Processes
ASTM E11	Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves
ASTM E70	Standard Test Method for pH of Aqueous Solutions with the Glass Electrode

(Copies of these documents are available online at http://www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 17025	General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document is available online at http://www.iso.org.)

SAE INTERNATIONAL

SAE-AMS-QQ-A-250/4	Aluminum Alloy 2024, Plate and Sheet

(Copies of these documents are available online at http://www.sae.org.)

2.4 Order of precedence.

Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification.

The engineered media furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.4).

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3.2 Safety data sheet (SDS).
An SDS shall be prepared and submitted to the independent testing laboratory in accordance with FED-STD-313.

3.3 Materials.
The finished product for plastic media shall be made from chlorine-free cured plastic stock of urea formaldehyde for Type II, melamine formaldehyde for Type III, or acrylic plastic for Type V, by processing to the desired size distribution specified herein. The finished product for Type VII engineered media shall be made from chlorine-free starch polymer by processing to the desired size distribution specified herein. The finished product for Type VIII engineered media shall be a Urea/Melamine Amino Thermoset plastic blend.

All finished product shall be magnetically cleaned prior to shipment.

3.3.1 Composition of plastic stock.
The finished plastic media product shall be manufactured from selected plastic stock of the exact chemical type required by this specification. Finished product shall be manufactured from virgin plastic stock (material in the form of unprocessed sheet or block made specifically for the production of plastic media) or from plastic stock formed from scrap plastic (material produced as a by-product of the manufacture of plastic items such as shirt buttons, dinnerware, etc.). Mixtures of different types of plastics or of plastic stock material shall not be permitted with the exception of Type VIII.

3.3.2 Fourier Transform Infrared Spectroscopy (FTIR) Attenuated Total Reflection (ATR).
All engineered media shall undergo Fourier Transform Infrared Spectroscopy (FTIR) Attenuated Total Reflection (ATR) analysis to confirm the material matches the media type description. The spectrogram of the finished product shall be essentially identical in shape (peak intensities may differ) to those on Figures 1 through 5, dependent on type, when analyzed as specified in 4.5.

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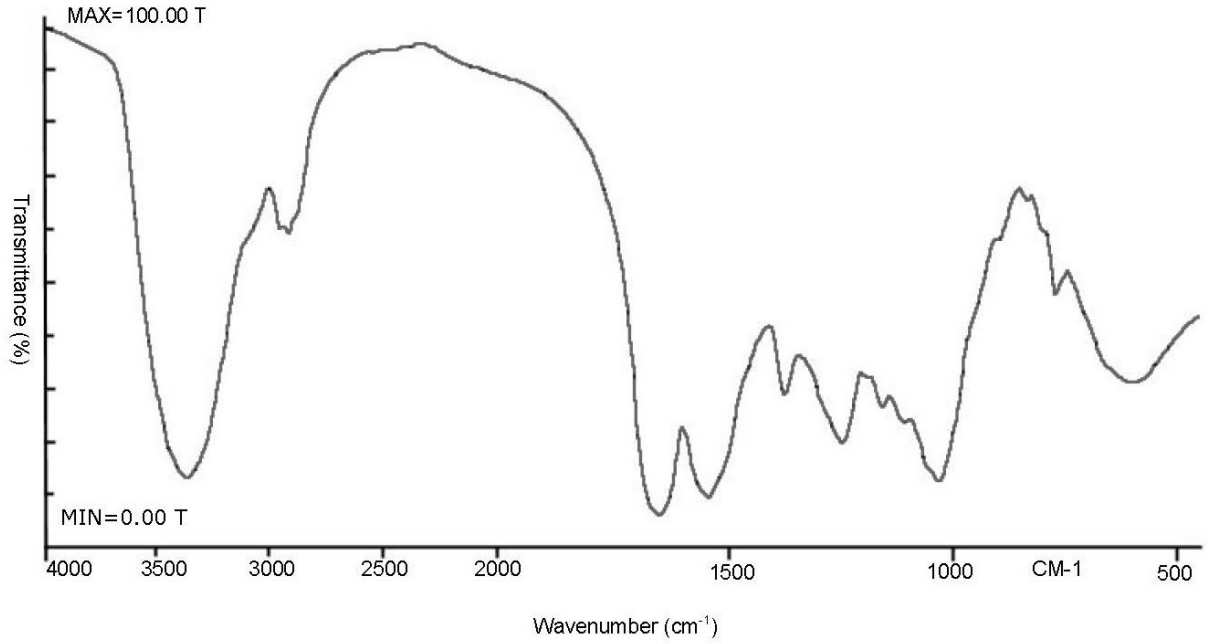


FIGURE 1. Type II urea formaldehyde.

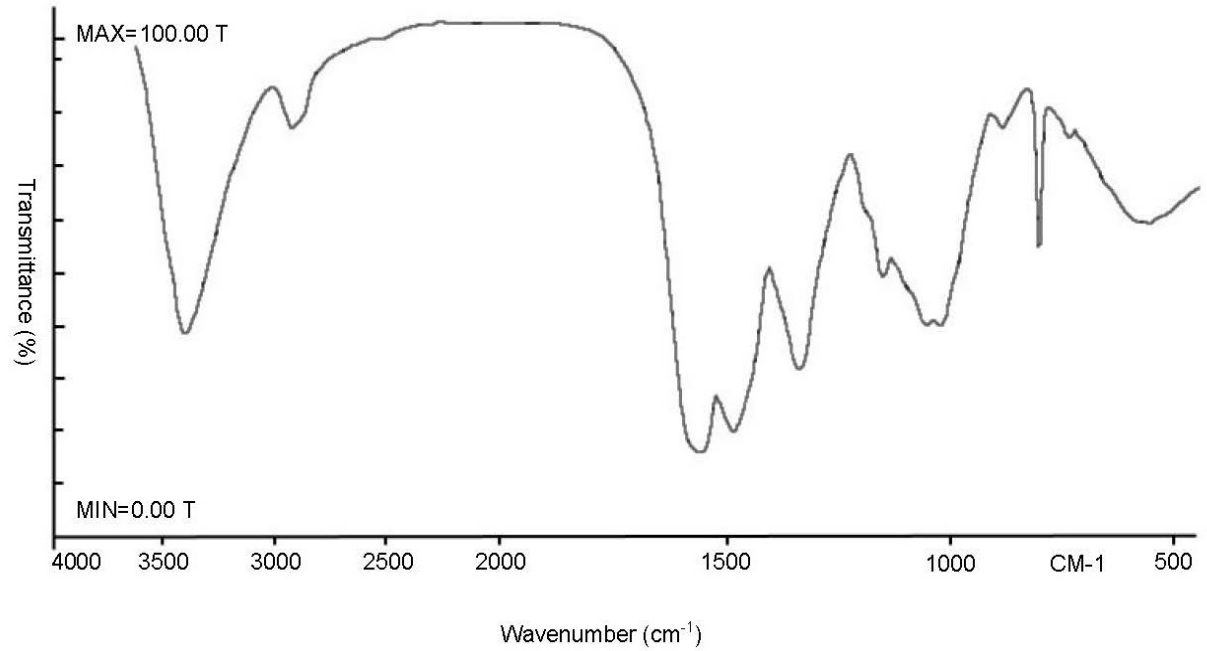


FIGURE 2. Type III melamine formaldehyde.

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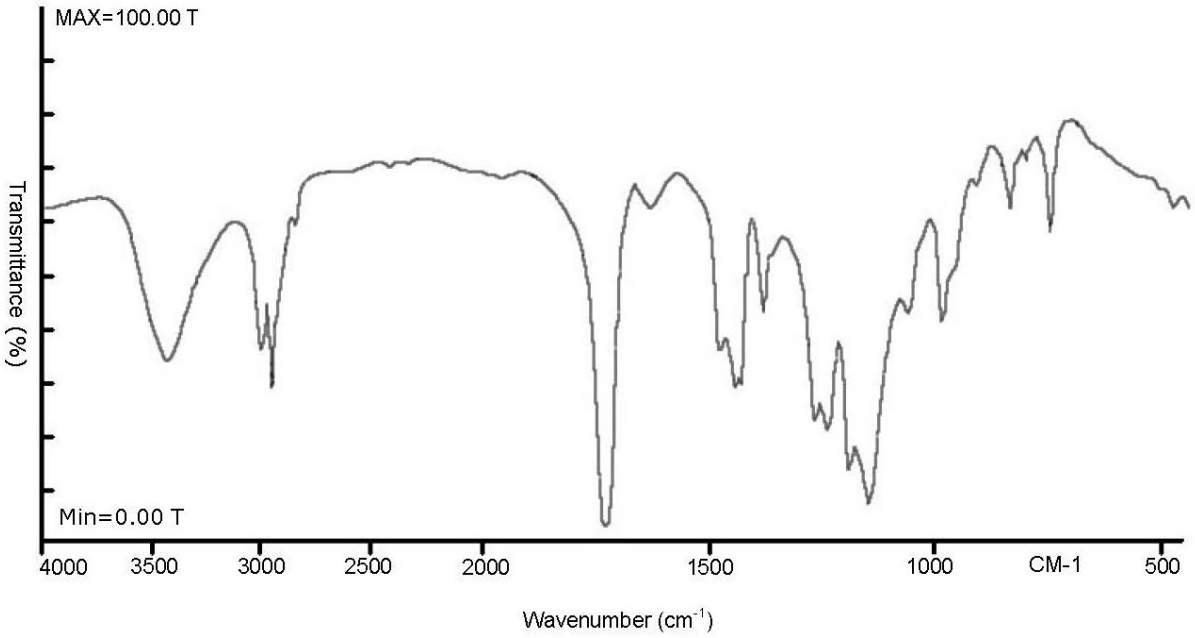


FIGURE 3. Type V acrylic.

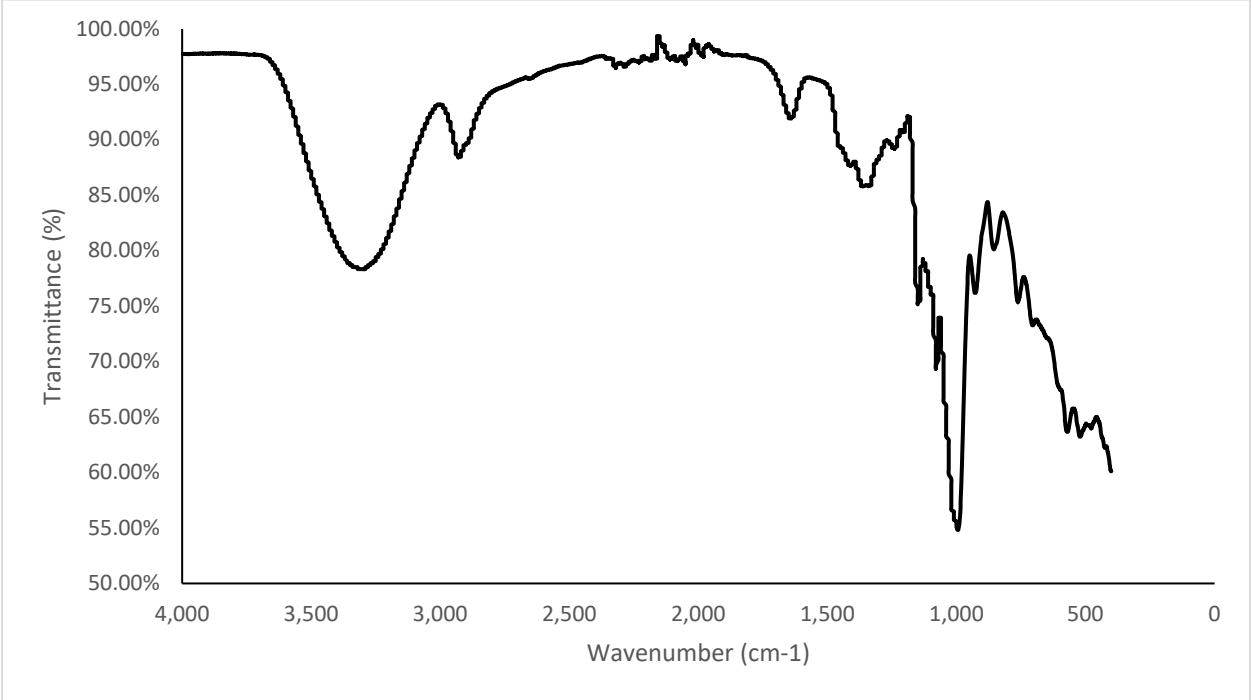


FIGURE 4. Type VII starch polymer.

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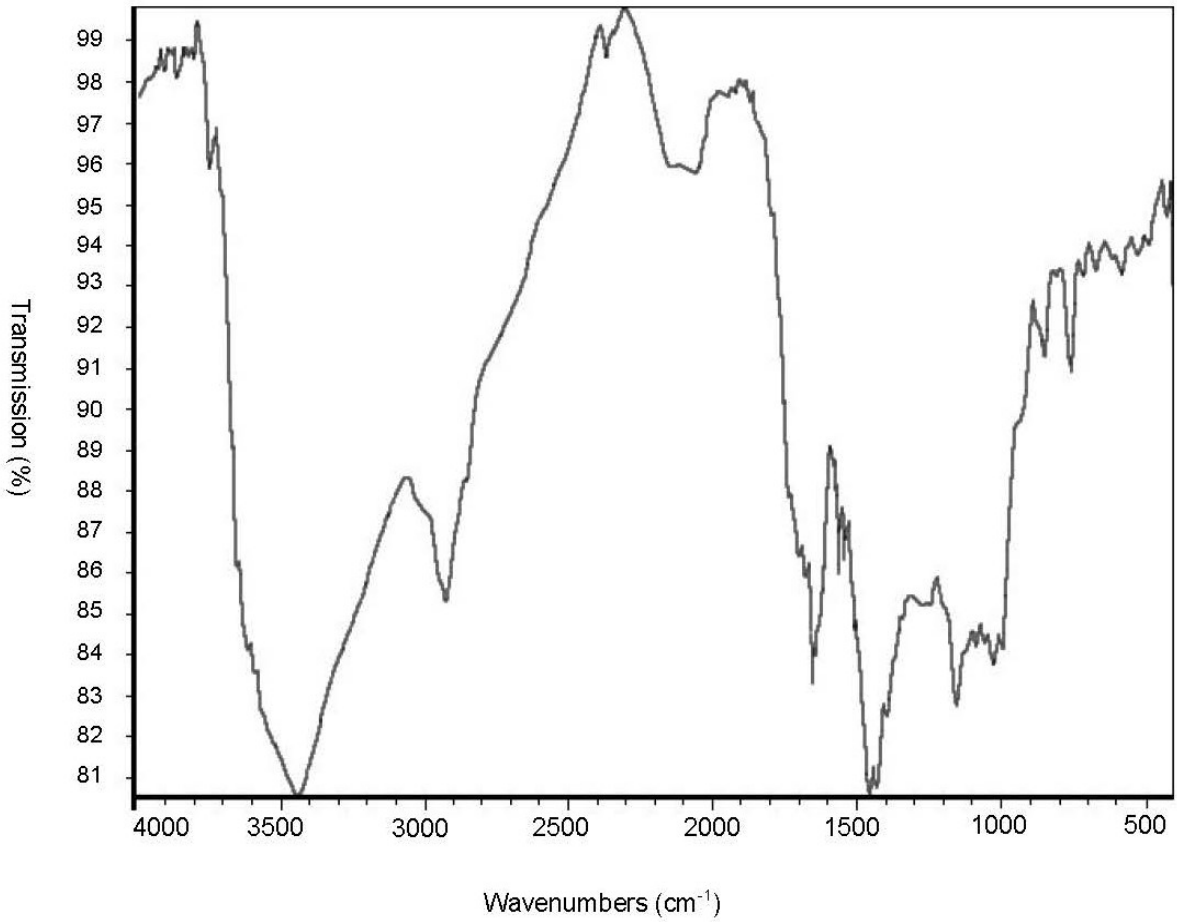


FIGURE 5. Type VIII Urea/Melamine multi-plastic blends.

3.3 Explosibility and ignitability.

The Naval Civil Engineering Laboratory (NCEL) Contract Report (CR) 87.011 of June 1987 covers an investigation of Types II, III, and V media. A similar evaluation of Types VII and VIII materials shall be required prior to procurement of that type in accordance with 4.6.

3.4 Toxicity.

The finished product shall have no adverse effect on the health of personnel when used for its intended purpose when verified in accordance with 4.7.

3.5 Physical and chemical properties.

The physical and chemical properties of the finished product shall be in accordance with Table I, when tested as specified in 4.8.

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TABLE I. Physical and chemical properties.

Property	II	III	V	VII	VIII	Test Para
Hardness	Barcol	Barcol	Barcol	Shore D	Barcol	4.8.1
minimum	54	64	46	72	54	
maximum	62	74	54	90	62	
Chlorine content, max	Trace	Trace	Trace	Trace	Trace	4.8.2
Ash content, max (% by wt.)	2.0	2.0	0.5	1.0	2.0	4.8.3
Iron content, max (% by wt.)	0.10	0.10	0.05	0.05	0.10	4.8.4
pH of Water Extract						4.8.5
minimum	4	4	4	4	4	
maximum	8	8	8	8	8	
Specific Gravity						4.8.6
minimum	1.47	1.47	1.10	1.38	1.36	
maximum	1.52	1.52	1.20	1.49	1.46	
Heavy particulates (% by wt.), max	0.02	0.02	0.02	0.02	0.02	4.8.7.2
Light particulates (% by wt.), max	1.0	1.0	0.1	1.0	1.0	4.8.7.3

3.6 Particle size.

The finished product shall have a mesh particle size distribution as specified in Table II when tested as specified in accordance with 4.9. The supplier shall identify the intended particle size for testing.

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TABLE II. Particle size distribution.

US Standard Screen Size	US Standard Particle Size Distribution																			
	12-20		12-16		16-20		20-30		20-40		20-50		30-40		30-60		40-60		60-80	
	Maximum %		Maximum %		Maximum %		Maximum %		Maximum %		Maximum %		Maximum %		Maximum %		Maximum %		Maximum %	
	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass	Retain	Pass
10	0.1		0.1																	
12	5.0		5.0		0.1															
16				20	15		0.1		0.1		0.1									
20		20		5.0		20	15		15		15									
25													0.1		0.1					
30		5.0				5.0		20		60		75	15		5.0					
35																	0.1			
40								5.0		20				20			5.0			
50												20							0.1	
60										5.0				5.0		20		20	5.0	
80																				20
100		1.0		1.0		2.0		2.0		2.0		2.0		3.0		5.0		5.0		5.0

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3.7 Surface residue.
Types II, III, VII, and VIII shall not produce a surface residue which interferes with the application of pretreatments when tested in accordance with 4.10. Blast residues from Types V shall be removable with methyl ethyl ketone.

3.8 Storage stability.
After storage for one year under normal storage conditions of a climate controlled environment at 21 ±8 °C (70 ±15 °F) and 50 ±10 percent relative humidity, the finished product shall meet all the requirements of this specification.

4. VERIFICATION

4.1 Classification of inspection.
The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 Qualification inspection.
Qualification inspection shall be performed on Aluminum Alloy 2024-T3 material conforming to SAE-AMS-QQ-A-250/4 as specified in each test.

4.2.1 Qualification samples.
Initial qualification will be completed by a laboratory approved by the United States Government, with request submitted to AFLCMC/EZPT-CPCO (see 6.4). Re-qualification will take place at either a lab approved by the United States Government or an independent third party lab in accordance with ISO 17025. Each sample shall be identified as follows:

Qualification test sample.

Specification MIL-DTL-85891.

Engineered Media, for Removal of Organic Coatings.

Manufacturer’s name and product type.

4.3 Conformance inspection.
Conformance inspection shall include tests in sections 4.5-4.10. Failure to meet any conformance test shall result in rejection of the batch and may constitute justification for removal of the product from the qualified products list. Rejected material shall not be resubmitted for acceptance without prior approval of the procuring activity. The application for resubmission shall contain full details concerning previous rejections and measures taken to correct the deficiencies.

4.4 Inspection conditions.
Unless otherwise specified, all inspections shall be performed in accordance with the test conditions in accordance with this specification.

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4.5 FTIR-ATR.
The finished product shall be prepared for analysis by grinding to a very fine powder with a suitable mortar and pestle (quartz or agate is recommended). The spectrum should then be obtained in accordance with the instructions for the FTIR-ATR.

4.6 Explosibility and ignitability.
An evaluation procedure follows: The finished product shall be tested in accordance with the U.S. Department of the Interior, Bureau of Mines recommended practice for determining such hazards as set forth in NCEL CR 87.011. Finished product which satisfies the following criteria are not considered explosive or ignitable hazards when used in a properly designed blast system:

Equipment	Media size	Parameter	Limit
1.2-L Furnace	200 mesh or smaller	Minimum auto ignition temperature (MAIT)	350 °C (662 °F)
20-L Chamber with 2500 J igniter	200 mesh or smaller	Lean flammability limit (minimum)	40g/cu M.
20-L Chamber with 2500 J igniter		Maximum explosible size	40 mesh

4.7 Toxicity.
Safety data sheets (SDS) for the dyes or pigments used in the media, the media itself, and for any material added to the media after curing (such as antistatic agents) shall be prepared and submitted in accordance with FED-STD-313. Questions pertinent to the effect of the finished product on the health of personnel when used for its intended purpose shall be referred by the acquiring activity to the appropriate medical service of the procuring agency as listed in FED-STD-313.

4.8 Physical and chemical property tests.

4.8.1 Hardness.
Barcol hardness (Types II, III, V, and VIII) shall be determined in accordance with ASTM D2583. Shore D hardness (Type VII) shall be determined in accordance with ASTM D2240. The hardness shall be determined on the selected stock before processing to the finished product. The hardness shall be in accordance with the requirement identified in Table I.

4.8.2 Chlorine content.
Weigh 0.05 gm of finished product into a disposable glass test tube. Add a piece of sodium metal (about 0.15 gram) to the tube, then heat over a Bunsen burner to a dull red. Allow the test tube content to cool, then add 0.50 mL ethanol to the tube and allow it to react until there is no evidence of residual sodium metal. Heat the test tube carefully to evaporate the ethanol and continue to heat until it glows a dull red. Immediately drop the test tube into 50 mL of distilled water, crush the lower end of the tube, and stir to dissolve any sodium chloride which has formed. Filter the solution into a 250 mL beaker, acidify with 10 or more drops of concentrated nitric acid, and add 10 drops of 0.1 M silver nitrate and swirl the beaker several seconds. The presence of chlorine is indicated by the development of a white precipitate which disappears upon the addition of several

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drops of concentrated ammonium hydroxide. The finished product shall be defined as having no more than a trace of chlorine when the white precipitate is less than that produced when 50 mL of a 16.5 ppm sodium chloride solution in distilled water is treated with acid and silver nitrate as stated.

4.8.3 Ash content.
Three grams of finished product shall be weighed to the nearest 0.1 mg into a tared ceramic crucible. The finished product shall be ignited with a fisher burner until no loss of material is visually apparent. The crucible shall be heated in a muffle furnace at 850 °C (1560 °F) to constant sample weight. The ash residue shall be calculated as a percent of initial sample weight.

4.8.4 Iron content.
Total iron content of the finished product shall be determined using the ash from 4.5.4 when tested in accordance with ASTM D3682. Total iron content shall be calculated as follows:

$$Total\ iron\ content\ (\%) = \frac{ash\ content\ (\%)\ X\ iron\ content\ (\%)}{100}$$

4.8.5 pH of water extract.
Twenty-five grams of the finished product shall be added to a clean 100 mL glass graduated cylinder that has been thoroughly rinsed with distilled water. Distilled water shall be added to the 100 mL mark of the cylinder, then stoppered. The cylinder shall be shaken for five seconds, then permitted to stand undisturbed for five minutes. The cylinder shall then be shaken again, per ASTM E70, and the pH of the mixture determined for conformance to the requirement in Table I.

4.8.6 Specific gravity.
Specific gravity shall be determined in accordance with ASTM D792 using n-butyl alcohol as the immersion liquid.

4.8.7 Heavy and light particulates.

4.8.7.1 Solvent preparation.
The solvent mixtures shall be prepared from n-hexane (specific gravity of 0.66) and a perfluorohexane solvent (3M PF-5060 Perfluorohexane, or equivalent) (specific gravity of 1.68). Two solvent mixtures shall be prepared to yield specific gravities as follows:

Mixture A for heavy particulates	1.40 for Type V.
	1.62 for Types II, III, VII, and VIII.
Mixture B for light particulates	0.10 less than specific gravity of finished product.

4.8.7.2 Heavy particulates.
Four hundred mL of Mixture A shall be added to 500 mL separator funnel, followed by the addition of 100 grams of the finished product. The funnel shall be swirled from side to side while in an upright position for one to two minutes, then be allowed to stand upright for five minutes. The settled material shall be drained from the funnel onto a 200 mesh sieve, tared to the nearest 0.001

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gram. The sieve shall be dried to constant weight in an oven maintained at 105 ±2 °C (221 ±4 °F). Heavy particulate contaminants shall be calculated as specified in 4.8.7.3.

4.8.7.3 Light particulate.
One hundred grams of the finished product shall be added to a tall form, 600 mL beaker, followed by 400 mL of Mixture B. The contents shall be stirred for one minute, then allowed to stand for five minutes. Any floating particulate shall be decanted onto a 200 mesh screen, tared to the nearest 0.001 gram. The sieve shall be dried to constant weight in an oven maintained at 105 ±2 °C (221 ±4 °F). The light contaminant particulate shall be calculated as follows:

$$Particulate\ contamination = \frac{Residue\ weight}{Sample\ weight}\ X\ 100$$

4.9 Particle size.
Particle size distribution shall be determined in accordance with ASTM D1921, Test Method B, and shall conform to the requirements in Table II. All screens used shall be qualified to ASTM E11.

4.10 Surface residue.
A 10 ×30 ×0.05 to 1.2 cm (4 × 12 ×0.02 to 0.50 inch) aluminum test panel conforming to SAE-AMS-QQ-A-250/4, T3 temper, shall be abraded manually with a very fine abrasive mat (see A-A-58054) wet with deionized water. The test panel shall be thoroughly rinsed with deionized water and wiped dry with bleached cheesecloth (see CCC-C-440, Class 1). One half of the aluminum test panel shall be masked with a similar panel. The exposed portion of the test panel shall be blasted for one minute with the finished product per TO 1-1-8 operation parameters. The blasted test panel shall be washed with a 10 percent by volume solution of MIL-PRF-85570, Type II, by swabbing gently with bleached cheesecloth. Panels blasted with Type V media shall also be wiped with cheesecloth wet with methyl ethyl ketone after the MIL-PRF-85570 wash. All panels shall be thoroughly rinsed with deionized water and wiped dry with the cheesecloth. The dried test panels shall be immersed in a chromate conversion coating solution (1 ounce, of MIL-DTL-81706, Type 1, Class 1A per 1 gallon deionized water) for three minutes. The panels shall be rinsed with deionized water for one minute. The blasted portion of the test panel shall be compared with the masked portion. A significant difference in color or non-uniformity of the MIL-DTL-81706, Type 1, Class 1A coating on the blasted portion of the panel indicates an interfering residue.

5. PACKAGING

5.1 Packaging.
For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.3). When actual packaging of materiel is to be performed by DoD or in-house personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Department or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

The engineered media covered by this specification is intended as an abrasive blasting material for paint removal. The requirements for engineered media as detailed in this specification are military unique to ensure media performance is acceptable when used IAW T.O. 1-1-8.

6.2 Acquisition requirements.

Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. Classification (see 1.2).
c Quantity required. (Used for procurement only (see 1.3)).
d. Packaging requirements (see 5.1).

6.3 Batch definition.

A batch is defined as a homogenous quantity of finished product, manufactured at one time or representing a blend of several manufactured units of finished product of the same formulation.

6.4 Qualification.

With respect to products requiring qualification, awards will be made only for products which are, at the time of the award of contract, qualified for inclusion in Qualified Products List QPL No. 85891 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from AFLCMC/EZPT-CPCO, 325 Richard Ray Blvd, Bldg. 165, Robins AFB, GA 31098-1640 or by email at afcorr@us.af.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

6.5 Subject term (key word) listing.

Environmental
Paint removal
Thermoset
Thermoplastic

6.6 Changes from previous issue.

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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CONCLUDING MATERIAL

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Preparing activity:
Air Force - 11

(Project 5350-2019-001)

Reviewers:
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